

Ozmoo

Ozmoo

A Z-machine interpreter for the Commodore 64 and similar computers

Release 15: 22 February 2026

Ozmoo was conceived and designed by Johan Berntsson and Fredrik Ramsberg. Special thanks to howtophil, Retrofan, Paul van der Laan, Jason Compton, Alessandro Morgantini, Thomas Bøvith, Eric Sherman, Paul Gardner-Stephen, Steve Flintham, Bart van Leeuwen, Karol Stasiak and Dennis Holmes for testing, code contributions and bug fixes.

Contents

Program and Data Structures	4
Overview	4
Source files	6
Startup	8
Z-machine	8
Disk I/O	9
REU	9
X16 builds	10
X16 Memory map	10
MEGA65 builds	12
MEGA65 Memory map	12
Save and Restore	14
Undo	15
Screenkernel	16
Version 6	17
The window model	17
Extended Color Mode on the C64	18
Full Colour Mode on the MEGA65	18
Per-window backgrounds and reverse video	20
Font 3	21
The VERA screen on the X16	21

The pictures screen	22
Pictures	24
What the X16 does differently	29
Mouse	30
The MEGA65's mouse	30
The X16's mouse	31
Clicks, and how they reach the game	31
The stack	32
Virtual Memory	37
The basics	37
Timestamps	38
The quick index	40
Efficient access to vmap	40
Banking	41
Continuous virtual memory optimization	42
REU Boost	42
Sound	43
Smooth Scrolling	44
Accented Characters	45
Assembly Flags	47
General flags	47
Internal Flags	53
Debug flags	54
Configuration blocks	56
Printer Support	58
Runtime Errors	59

Program and Data Structures

Overview

The main parts of Ozmoos are as follows:

- The interpreter This is the main program that sets up the memory structures, reads the config into memory, and then starts executing the program instruction by instruction in the Z-machine emulator, taking care to read portions of the Z-code file from disk as needed.
- Virtual memory buffers On some computers it is hard to use all the available RAM without time-consuming banking operations. To improve efficiency the virtual memory buffers are used when Ozmoos is compiled with virtual memory support, and located in RAM that is guaranteed to be accessible. By having the buffers in accessible RAM most operations can be done with direct RAM access, and banking is only needed when the program counter leaves the buffer. This system is used on C64 and C128.
- Z-machine stack This is used to store arguments and function calls. The stack is described in detail in a chapter of its own.
- Virtual memory The rest of the available RAM is divided into 512-byte blocks and used to store parts of the story file as needed by the game state. The virtual memory is a complex topic which is covered in more detail in a chapter of its own.

The next figure shows how memory may be allocated when playing a game on the Commodore 64.

Memory map for Ozmoo, playing a Z5 story

2018-10-15

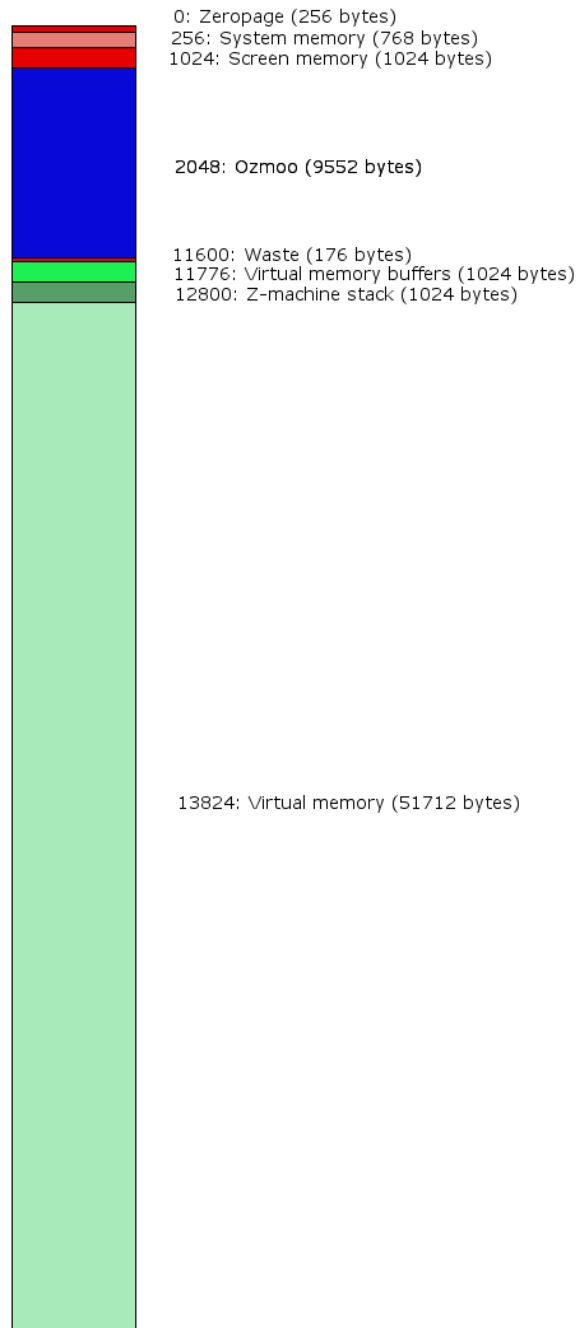


Figure 1: C64 memory overview

Source files

These source files are located in the asm dictory. The table also shows the most important routines included in each source file.

File/Routine	Comment
ozmoo.asm <i>program_start</i>	Ozmoo's main routine where execution of Ozmoo begins
zmachine.asm <i>z_execute</i>	implements the Z-machine execute the Z-machine program
c65toc64wrapper.asm	code to go to C64 mode on the MEGA65
constants.asm	zero page allocations and kernal labels
constants-c128.asm	C128 zero page allocations and kernal labels
constants-header.asm	labels to access the header part of a story file
dictionary.asm	routines to access the dictionary of a story file
disk.asm <i>readblock</i> <i>readblocks</i> <i>read_track_sector</i>	read and write to floppy disk read a memory block from the story file read a memory block from the story file read single sector from the floppy disk
file-name.asm (in temp folder)	holds the name of the boot file (file is generated by make.rb)
file-name.tpl	template used by make.rb to generate file-name.asm
memory.asm	routines to access the Z-machine memory
objecttable.asm	routines to access the object tree defined in a story file
picloader.asm	code to show a picture while reading the story file
pictures-mega65.asm	the version 6 picture engine for the MEGA65: the Full Colour Mode tile store, and decrunching the pictures' Exomizer archive into Attic RAM at boot

File/Routine	Comment
pictures-x16.asm	the version 6 picture engine for the X16: VERA's tile layer, and loading a picture from SD when it is drawn
reu.asm	implements the Ram Expansion Unit interface for C64 and C128, banked RAM for the X16, and an interface to Attic RAM on MEGA65
screen.asm	printing routines
screen-z6.asm	printing routines for version 6, with its window model and graphics
screenkernel.asm	low-level display routines
screenkernel-z6.asm	low-level display routines for version 6
sound.asm	routines to support playing sound effects
sound-aiff.asm	routines to support sound effects in AIFF format
sound-wav.asm	routines to support sound effects in WAV format
splashlines.asm (in temp folder)	contains splash screen text (file is generated by make.rb)
splashlines.tpl	template used by make.rb to generate splashlines.asm
splashscreen.asm	show the splash screen
stack.asm	implements the Z-machine stack
streams.asm	implements text streams for the Z-machine
text.asm	routines to read and write text
utilities.asm	various utilities, error handling and debug routines
vdc.asm	low level routines to write text on the C128 in 80 column mode
vera.asm	defines the interface to the VERA chip for text and graphics output on the X16

File/Routine	Comment
vmem.asm	virtual memory routines, and corresponding routines for non-vmem builds
walkthrough.asm (in temp folder)	contains walkthrough commands for benchmarking (file is generated by make.rb)
walkthrough.tpl	template used by make.rb to generate walkthrough.asm
zaddress.asm	routines to access data (i.e. not program code) in the Z-machine memory space

Startup

Ozmoo programs are compressed with Exomizer into a file called **story**, which is loaded and run like a Basic program. This file is sometimes referred to as a boot file. When this program is executed, the uncompressed Ozmoo program replaces **story** and execution starts from the `program_start` label.

First the screen is initialised and the splash screen displayed (if any). Then the config blocks (the first two blocks on the boot disk) are read, Ozmoo checks for an REU, the interpreter may ask the player to insert the story disk(s) into the drive(s), Ozmoo may read all of the story file into the REU, or it may load parts of it into RAM (if there is still some empty RAM after uncompressing the boot file). Then the Z-machine program counter is set to the start of the story, as specified in the header, and execution of the Z-machine program begins.

Z-machine

The Z-machine executes instructions, using the Z-machine stack to keep track of calls and arguments. A Z-code file (AKA game file or story file, like `zork1.dat` or `planetfall.z3`) is essentially a snapshot of the memory of a virtual computer (the Z-machine). A Z-code file is divided into dynamic memory (up to 64 KB) and static memory (everything above dynamic memory, up to 512 KB). Dynamic memory is the only part that can be altered during program execution. Ozmoo keeps the entire contents of dynamic memory in RAM at all times, while using whatever RAM is left for virtual memory, to hold as much of static memory as possible. Whenever the player moves into a new location or tries something new and the disk starts spinning, this is because the program needs a part of static

memory that isn't in RAM, so it has to be fetched from disk. What's so nice about this system is that it's handled entirely by the interpreter. The author of a game doesn't have to figure out when to load which parts of the story file - they can just write their game as if it's going to run on a computer with 128 or 256 KB of RAM, and the interpreter makes it work on a computer with much less RAM than that.

To further conserve space, the Z-machine has a text compression scheme built in. Basically it means each character is translated into one or more five-bit codes. By default, the space character and all lowercase letters fit into a single five-bit code each, while uppercase letters and special characters require two or four five-bit codes. These five bit codes are then grouped together so three of them occupy two bytes. I.e. "adventure" is stored as "adv" in two bytes, "ent" in two bytes and "ure" in two bytes, thus using six bytes. On top of this, there is a system of abbreviations, so common strings can be replaced by two five-bit codes.

Disk I/O

Ozmoo is designed to use a floppy disk layout similar to what Infocom used, but it has some extra options such as using multiple disk drives to play bigger games. Config information for the game is stored in sector 0 and 1 of track 1 on the boot disk. The static memory portion of the story file is stored in the track 1 and up on the story disk(s). The boot file, containing the interpreter, dynamic memory and as much of static memory as possible, is stored on the boot disk or in the remaining free space of the combined boot / story disk.

The disk I/O routines are located in disk.asm. The virtual memory uses read_block and read_blocks to read data from the story file when needed. There is a routine read_track_sector for reading a single sector of the story file specified by track number and sector number.

REU

If Ozmoo detects a RAM Expansion Unit (REU), and it's big enough to hold the entire story file, the player gets the question if they want to use it. If they do, the static part of the story file is read into the REU before the game starts. When the virtual memory system needs to read data from the story file by calling read_block or read_blocks (disk.asm) it will read the blocks from the REU instead of the floppy drive.

The interpreter detects the size of the REU in banks, where a bank is 64 KB. After deciding whether to use the REU for caching the story file, it may also allocate a bank for undo and a bank for scrollbar buffer, if there is room and

the game was built with support for these features. If the story file is stored in REU, it always begins in bank 0 and may use up to 8 banks. If story data is stored in the REU, the first page of bank 0 is reserved for fast page copying operations. If story data is not stored in the REU, and the REU is 256 KB or bigger, the first page of bank 3 is used for fast page copying operations instead. If the REU is smaller, fast page copying isn't used.

X16 builds

On the X16, the main game file doesn't hold any story data. The entire Z-code file is held in a file called "zcode" and is loaded into banked RAM on boot. Since the entire file is held in one linear chunk of memory, and accessed in place (using banking), this is not a virtual memory build, and the VMEM constant is not defined. No config information is needed, so there are no config blocks on disk. This means a X16 game can be copied from one disk to another using a regular file copier.

A version 6 game built with Z6_PICTURES also has one file per picture, named [P001] up to [P999], sitting in the game's directory beside [ZCODE]. Unlike the MEGA65's, they are not compressed and are not preloaded: a picture is read from the SD card the first time the game draws it. See "Version 6".

X16 Memory map

Similar to C128, the graphics output is handled by a chip called VERA that contains the video memory. This means that no normal memory is needed for screen RAM. There is a standard 512 KB extended memory, of which a chunk/bank at a time is available at \$a000-\$bfff. The currently accessible bank of the extended memory is controlled by setting \$0000 to the chunk number.

Normal memory

Address range	KB	Usage
\$0000-\$7eff	32	System RAM, interpreter, z-stack
\$7f00-\$9eff	8	Story file, low-RAM slice (from X16_STORY_BASE)
\$a000-\$bfff	8	Story file, banked chunk (bank 1 and up)

The story's first pages live in normal RAM from X16_STORY_BASE (\$7f00); the rest is reached through the banked window at \$a000. The interpreter and the z-stack must end below X16_STORY_BASE, and three sites map a z-page to

a bank with the same 8 KB-aligned arithmetic ($X16_LOW_STORY_PAGES + (bank - 1) * 32$), so the constant is required to be a multiple of 32 pages. The slice was 16 KB at \$5f00 until the picture builds needed the room; dropping it to 8 KB freed 8 KB of interpreter space at the cost of one more story bank of the 512 KB.

Video memory (VERA)

The standard text mode needs $80 \times 60 \times 2 = 9600$ bytes, starting from \$1b000, since the screen is 80 columns, 60 rows, and each character is stored as a (character code, colour) tuple.

Address range	KB	Usage
\$1b000-\$1d580	about 10	Screen RAM

A version 6 build with `Z6_PICTURES` puts the pictures on a second VERA layer behind the text, and then most of the 128 KB of video RAM is in use:

Address range	KB	Usage
\$00000-\$0fff	64	Picture tile store: 1024 tiles of 64 bytes (<code>Z6_PICTURES</code>)
\$10000-\$10fff	4	Layer 0 tile map, 64 x 32 entries of two bytes (<code>Z6_PICTURES</code>)
\$11000-\$117ff	2	The character set, moved out of the tile store's way (<code>Z6_PICTURES</code>)
\$13000-\$13fff	4	Sprite 0's image: the kernal's mouse pointer (<code>Z6_MOUSE</code>)
\$1b000-\$1c380	4	Screen RAM (80 x 25 in a pictures build)
\$1fa00-\$1fbff	0.5	Palette RAM: 256 entries of two bytes
\$1fc00-\$1ffff	1	Sprite attributes

Note that in an ordinary build the character set is left where the kernal keeps it, in the first 64 KB, which is exactly where the tile store has to go — hence the copy to \$11000 and the repointing of layer 1's tile base. Ozmoo also stops using video RAM for the undo buffer in a pictures build, for the same reason; see “Undo”.

Extended/banked memory

Address range	KB	Usage
\$00000-\$02000	8	Reserved space
\$02000-\$7fff	504	Story file

The extended memory is mapped into \$a000-\$bfff, so the memory is accessible in 8KB chunks. Note that the first 8KB is used by the kernal and DOS routines, so we load the story file from \$02000.

In a version 6 build with Z6_PICTURES, two more regions are carved out of the banked memory above the story. The picture being drawn is read into a staging area of four banks (32 KB, which is more than the largest picture file in any of the four version 6 games), and if undo is enabled its buffer follows above that. Both start from the first bank the story does not use, which make.rb computes from the story’s size and passes in as PIC_STAGING_BANK; it refuses the build if the three together would not fit in the 512 KB. With the largest of the four games, Shogun’s 345 KB, the story ends in bank 41 and there is room to spare.

MEGA65 builds

On the MEGA65, the boot file doesn’t hold any story data. The entire Z-code file is held in a SEQ file called “zcode” and is loaded into Attic RAM on boot. Since the entire file is held in one linear chunk of memory, and accessed in place, this is not a virtual memory build, and the VMEM constant is not defined. No config information is needed, so there are no config blocks on disk. This means a MEGA65 game can be copied from one disk to another using a regular file copier.

A version 6 game built with Z6_PICTURES also carries its pictures, as one Exomizer archive per picture disk, decrunched into Attic RAM at boot. These archives live on their own picture disk(s), not the boot disk; every shipped game fits a single one. See “Version 6”.

MEGA65 Memory map

Address range	KB	Usage
\$00000-\$ffff	64	System RAM, screen RAM, interpreter Picture tile store (version 6, Z6_FCM_MODE without pictures); current sound effect (Z6_PICTURES)
\$10000-\$1fff	64	
\$40000-\$40fff	4	Screen RAM for scrollbar mode

Address range	KB	Usage
\$40000-\$4ffff	64	Current sound effect; picture tile store (Z6_PICTURES)
\$50000-\$5ffff	64	Undo buffer; picture tile store (Z6_PICTURES)
\$08000000-\$0807ffff	512	Story file
\$08080000-\$080800ff	0.25	Signature of last game that was loaded (to make restart faster)
\$08100000-\$081fffff	1024	Sound data
\$08200000-\$082fffff	1024	Scrollback buffer
\$08300000-\$083000ff	0.25	Picture restart signature (version 6, Z6_PICTURES)
\$08300100-		Pictures, decrunched (version 6, Z6_PICTURES)
\$08510000-		One disk's crunched archive, staged while it is decrunched (Z6_PICTURES)
\$08600000-		Undo buffer (Z6_PICTURES)
\$0ff80000-\$0ff807ff	2	Colour RAM
\$0ff80800-\$0ff817ff	4	Colour RAM for scrollback mode

Bank 1, \$10000-\$1ffff, is untouched in an ordinary build: nothing in Ozmo addresses it, and the C64 mode kernal leaves it alone. A version 6 build in Full Colour Mode uses it as the tile store, which holds 1024 tiles of 64 bytes.

That is enough for a full 40x25 screen in which no two cells are alike (the 80-column screen's 2000 cells would already outgrow it), but not for a game that keeps several pictures on the screen at once. Arthur draws its interface as pictures — a border, a scene and a status panel are all live together, over a thousand tiles — so a build with Z6_PICTURES moves the store to \$40000-\$5ffff, giving 2048 tiles, and moves the two things that lived there out of the way. The current sound effect goes down into bank 1, which the tile store has vacated; it has to stay in fast RAM because the audio DMA's stop address is only 16 bits wide, so a sample must sit at the foot of a bank. The undo buffer goes to Attic RAM at \$08600000, well clear of the decrunched pictures, because it is only ever reached by DMA and so can live anywhere. Banks 2 and 3, \$20000-\$3ffff, are the ROM and can never be used for any of this: the C64 font the Full Colour Mode text is drawn with is read from \$2d800.

A single picture still has at most 2000 unique tiles, since it covers at most the 1000 logical cells of the screen, each of which is two tiles on the 80-column screen.

In Full Colour Mode a cell is two bytes in both screen RAM and colour RAM. On the 80-column full colour screen that is 4000 bytes each: the screen grows to \$0800-\$179f, ending just under the program at \$1800, and the colour bytes outgrow the 2 KB window at \$d800 altogether — which is why all colour access

under `Z6_FCM_MODE` goes through 32-bit pointers to colour RAM's real home at `$ff80000` instead of `colour2k/colour1k`. The legacy 40-column screen (`-fcm:40`) uses the same 2000 bytes as the 80x25 text screen.

Note that the scrollbar buffer's screen RAM and the current sound effect both begin at `$40000`. Opening the scrollbar while a sound is playing therefore corrupts the first 4000 bytes of the sample. It repairs itself, because the next `@sound_effect` copies the sample from Attic RAM again. Version 6 games never have a scrollbar buffer, so this does not collide with the tile store.

Save and Restore

The story file is read piece by piece by mapping the program counter to the correct track and sector, and read it into memory. The main function doing this is `readblocks` located in `disk.asm`

`disk.asm` also contains save and restore functionality. The main functions are `do_save` and `do_restore`. The save files are normal files that contain some important internal variables such as the program counter, the Z-machine stack, and the `dynmem` part of the RAM.

Undo

If extra memory is available then it can be used to support undo for games that probe for this functionality. This is currently supported for C64 and C128 with a RAM Expansion Unit (REU), and for the MEGA65. To enable undo support the `make.rb` script should be called with the `-u` switch. If undo is enabled and the header has the undo flag set, then Ozmoos checks for available memory. If not found, then the undo header flag is cleared, and an error message is shown.

When undo support is active then the current game state (the stack, dynamic memory and some variables) are copied into the undo buffer before a new command is read from the user. This is similar to the save command, but saves to memory instead of saving to a file. If the user gives the undo command, then the state (dynamic memory, stack etc.) is instead copied from the undo buffer, and the execution continues from the saved state, undoing the previous turn.

The main functions for undo are `do_save_undo` and `do_restore_undo` in `disk.asm`. As in Frotz, a hotkey (Ctrl-U) has been added to enable undo support for z3 games, even if official undo support is only available for z5 and upwards. The hotkey handling code is found in `text.asm`.

On the MEGA65 the undo state is moved by DMA, so the buffer can live anywhere. Its address is the pair of constants `UNDO_BANK` and `UNDO_ADDRESS_TOP`: bank 5 of fast RAM normally, and Attic RAM at `$08600000` in a version 6 build with pictures, where the tile store takes bank 5 over.

On the X16 the undo state normally goes into video RAM, which is otherwise unused below the text screen. A version 6 build with pictures fills that RAM with the tile store, so the buffer moves into banked RAM instead, just above the picture staging area (`PIC_UNDO_BANK`). Since source and destination then share the one `$a000` window, `do_save_undo` and `do_restore_undo` copy through a pair of helpers that switch the RAM bank around every byte and hand the caller's bank back.

Screenkernel

Screenkernel, implemented in `screenkernel.asm`, is a replacement for the low-level screen output routines on the Commodore computers. It is needed to abstract away low-level implementation details so that new targets can be more easily added, and also to support custom text scrolling to efficiently enable status lines, especially big ones used in games such as *Border Zone* and *Nord and Bert*. The number of lines to protect when scrolling is stored in `window_start_row`

The main functions of screenkernel are `s_printchar`, which replaced `CHROUT $FFD2`, and `s_plot` which replaced `PLOT $FFF0`. It also provides `s_set_textcolour`, `s_delete_cursor`, `s_erase_window` `s_erase_line`, `s_erase_line_from_cursor` and `s_init`.

Screenkernel is started by calling `s_init`. Internally it keeps track of the cursor position so it can put a character on the screen when `s_printchar` is called.

For the Commodore 64 version the characters are stored directly in the video memory, and the colour in the colour memory. The Commodore 128 version detects if 40 or 80 columns mode is used while running the program. If 40 characters are used, then it works like the Commodore 64 version. But if 80 columns are used, then the C128's Video Display Controller chip (VDC) is used. Instead of writing directly into the video memory, character output, scrolling and other screen commands are sent by VDC registers. The file `vdc.asm` contains functions that make this communication easier.

The X16 is using a separate chip called VERA for text output, similar to VDC for the C128. The file `vera.asm` defines the interface to the VERA chip used for putting characters on the screen.

The Plus/4 and Commodore 128 in 80 column mode doesn't use the same palette as the Commodore 64. Mapping tables (`plus4_vic_colours` and `vdc_vic_colours`) are used to assign C64 colours to their closest equivalents on these platforms.

Version 6

Version 6 of the Z-machine replaces the two windows of versions 3 to 5 with eight, each with its own rectangle on the screen, its own cursor and its own colours, and it expects the interpreter to draw pictures. Ozmoos implements the window model on every target, and the pictures on the MEGA65 (in Full Colour Mode) and on the X16 (on a VERA tile layer). The C64 and the Plus/4 will never draw pictures.

Because so much of the screen layer changes, version 6 gets its own copies of the two screen files: `screen-z6.asm` and `screenkernel-z6.asm`, sourced instead of `screen.asm` and `screenkernel.asm` when the Z6 flag is set. They are meant to stay close to the files they were forked from; when the originals change, it is better to fork them again than to patch the copies by hand.

The window model

The eight windows are held in the property arrays the standard describes (`window_y`, `window_x`, `window_y_size`, `window_x_size`, and so on), laid out one after another so that `get_wind_prop` and `put_wind_prop` can find a property with `window_y + 8 * property + window`. Coordinates are stored 0-based; the opcodes convert to and from the 1-based coordinates the Z-machine uses.

Property 13, the font size, is the only property that is a real 16-bit word: the height in the high byte and the width in the low one. It cannot live in the byte-per-window arrays with the others, so `get_wind_prop` answers it directly, with 1 and 1, because the header tells the game that a character is one unit wide and one unit high. A game may divide by it, so it must never be zero.

Printing, wrapping, scrolling, the cursor, the MORE prompt and erasing are all per window: text wraps at the window's right margin, a window scrolls its own rectangle, and the MORE prompt appears in the window's bottom right cell rather than the screen's. Each window keeps its own cursor and its own line count.

Every target scrolls a window rather than the screen, and they all do it with the same code: `.calc_window_rect` works out the window's rectangle, `.sw_up_one`

and `.sw_down_one` move its rows, and `.sw_blank_row` blanks the one scrolled into view. Only the row copy is per machine. The C64, the Plus/4 and the MEGA65 have their screen in the CPU's address space and copy it themselves; the X16's VERA and the C128's VDC do not, and get a row copy of their own (`.vera_copy_row` and `.vdc_copy_row`).

The VDC's is worth a word, because the chip has a blitter and it would be a waste not to use it. With the copy bit of the vertical scroll register (24) set, writing a byte count to `VDC_COUNT` (30) copies that many bytes from the copy source address (registers 32 and 33) to the current address (18 and 19). A row of a window is therefore one register write rather than eighty read-and-write round trips through the data port; the characters at \$0000 and their attributes at \$0800 are copied in turn. A count of 0 means 256 bytes rather than none, so a window with no columns has to be turned away before the blitter is asked to copy a quarter of the screen. This is the same blitter the old whole-screen scroll used, driven a row at a time between two arbitrary rows instead of down the whole display.

Extended Color Mode on the C64

The C64 has one background colour, which is no use to a game that wants eight windows in eight colours. `Z6_ECM_MODE`, selected with `make.rb`'s `-ecm` switch, turns on the VIC-II's Extended Color Mode: the top two bits of a screen code choose one of four background registers, \$d021 to \$d024, so each window can have its own background. The price is that only 64 characters remain, so screen codes are masked to six bits, uppercase renders as lowercase, and reverse video is not available. ECM is C64 only; `make.rb` rejects it on other targets.

Full Colour Mode on the MEGA65

The MEGA65 does not need the ECM compromise. `Z6_FCM_MODE`, selected with `-fcm`, puts the VIC-IV into Full Colour Mode with 16-bit character codes on an 80x25 screen over a 640x200 (H640) canvas. The version 6 games assume 80 columns — Zork Zero's layout breaks and Journey is simply unplayable at 40 — while their art is drawn to a 320-wide scale, so the text uses the ROM font's ordinary 8-pixel glyphs and the pictures are pixel-doubled as they are baked (see Pictures), keeping their intended size. `-fcm:40` (`Z6_FCM_40`) still builds the old 320x200, 40x25 screen; it renders text identically to the C64, which makes it the line-for-line regression companion, and its disks carry an `__fcm40` suffix so the widths never share artifacts.

The registers are set up in `init_mega65`, with the hot registers turned off so the values stick. The important ones are \$d031, where H640 is set (cleared

under `-fcm:40`), `V400` is cleared and, less obviously, `ATTR` must also be cleared, because that is what selects 8-bit colour; and `$d054`, where `CHR16` and `FCLRHI` are set while `FCLRLO` is left clear. `FCLRLO` clear is the whole trick: screen codes below 256 are still ordinary 8-byte glyphs fetched from `CHARPTR`, so all the text renders as before, while codes from 256 up are 64-byte full colour tiles. `CHARPTR` points at `$2d800`, the C64 font in the MEGA65's ROM, whose codes 128 to 255 are the reversed glyphs, so reverse video keeps working. The font is the usual 4 KB in two halves, uppercase and graphics first and mixed case second, exactly as on a C64; `$2d800` is the second half. Pointing it at `$2d000` instead renders lowercase as capitals and capitals as graphics, which is invisible in an all-lowercase test game and obvious the moment a real one prints a sentence.

A cell is two bytes in both screen RAM and colour RAM, so a row of 80 cells is 160 bytes and the two buffers are 4000 bytes each. The screen still fits below the program (`$0800-$179f`), but the colour bytes no longer fit the CPU's 2 KB window at `$d800`, so under `Z6_FCM_MODE` every colour access goes through a 32-bit pointer into colour RAM at `$ff80000` (`sta [zp],z`) instead of the old colour2k/colour1k banking. `zp_colourline` is four bytes for it, at `$e5` — not at its old `$f3`, because the kernal's keyboard scan rewrites `$f5/$f6` with its decodable pointer on every IRQ and would clobber the pointer's high word; `$d9-$f2` is the kernal screen editor's line-link table, which Ozmoos's own screen code replaces, so that region is interrupt-safe and also houses the two scroll scratch pointers and the `[More]` cell's pointer. Ozmoos writes the character into the even byte of a screen cell and the colour into the odd byte of a colour cell. The other two bytes, the screen cell's high byte and the colour cell's flags, must be zero and stay zero: `init_mega65` clears them once.

The code that reaches every cell would otherwise have to change everywhere, so:

- `zp_screencolumn` still holds a column, and only the sites that index the screen double it. That leaves the window edges, the margins and every comparison alone.
- `zp_colourline` is biased by one, so the same doubled index writes both the character and the colour. A row offset's low byte is a multiple of 32, so the bias never carries.
- Every place that writes a character also writes zero to the cell's high byte, with the `clear_cell_high_byte` macro. Without it, a character printed over a picture would leave the cell pointing at a tile.

Note that most text does not pass through `s_printchar` at all: it goes through `print_line_from_buffer` in `screen-z6.asm`, which writes the screen directly.

Because `s_printchar` only ever writes two-byte FCM cells, there is no separate text mode to fall back to, so the splash screen renders in Full Colour Mode like everything else. Its lines are centred for 40 columns and get the same `+20`

re-centring as the 80-column text build; only -fcm:40 skips it, where +20 would push them off the right edge and wrap into garbage. On quit, `z_ins_quit` clears `CHR16` and `FCLRHI` again – the C64 reset it then performs never touches `$d054` – and turns the mouse sprite off, so the machine returns to a legible BASIC screen rather than a 16-bit-character full colour one.

Per-window backgrounds and reverse video

A version 6 window has its own colour pair (property 11: the background in the high nybble, the foreground in the low). `set_colour` keeps each window's pair in `window_colour`, and `get_wind_prop` answers it, because the games read it back — Arthur reads the pair and sets its exact swap to box a parser message the way it boxes the status line.

How that pair reaches the screen depends on what the hardware can do:

- **The X16 has a real per-cell background.** VERA's colour byte carries the background in its high nybble, so under `Z6_WINDOW_BG` (set for the X16) each window simply renders its own background colour. `x16_apply_window_colour` points the live VERA colour (background high nybble, foreground low) at the current window's pair; it runs on `set_colour` and on every window switch, so a window always prints in its own colours. Because the colour travels with the cell, the background fills on erase and survives a scroll for free. This is the clean version of what ECM does with four registers, and it retires the reverse-video fake below on the X16.
- **Every other target fakes it with reversed glyphs.** The C64, the C128, the Plus/4 and the MEGA65's full colour screen have no per-cell background, so a *swapped* pair — one whose foreground equals the screen background, which is exactly what the games ask for — is drawn as reversed glyphs: the field takes the requested background colour and the text takes the screen background. The glyphs already fill and scroll a cell at a time, but the swap *state* is per window, not global: `window_swap` records each window's, and `apply_window_swap` restores it (with the field colour) on a window switch, so a reversed field no longer bleeds onto the next window or comes out the wrong way round after a scroll. It forces the colour only for a swapped window, whose field colour is an explicit choice; a non-swap window keeps the global (and darkmode) colour.
- ECM keeps its four-register scheme (above); the reversed-glyph fake and `Z6_WINDOW_BG` both stand down on that target.

Font 3

Font 3 is the standard's character-graphics font (section 16), a set of box-drawing glyphs. Ozmoos has no such charset, but the only version 6 use of it is Journey's command-menu dividers — the four corners, horizontal and vertical lines, and a solid block that marks the connected command. Journey decides font 3 is available from the interpreter number, not from `set_font`'s result, so it prints the glyph codes regardless; on a plain interpreter they used to come out as stray punctuation and digits.

`set_font` now accepts font 3 and records it per window (`window_font`, property 12; the shared `z_font` is only two bytes, but a version 6 window runs 0 to 7). When the current window is in font 3, `font3_translate` maps each glyph code to the equivalent PETSCII box-drawing character before it enters the print buffer, so the ordinary `convert_petscii_to_screencode` path renders it natively on every text target — the C64, the C128's two screens, the Plus/4, the MEGA65 full colour text screen and the X16 — with no per-target code. Codes the games do not use pass through unchanged.

The VERA screen on the X16

The X16's screen is not in the CPU's address space at all: VERA holds it, and it is reached through a pair of address-and-data port pairs. A text cell is two bytes, a character code and a colour, and a text row is 256 bytes whatever the screen's width, so a cell lives at $\$1b000 + \text{row} * 256 + \text{column} * 2$. That is convenient: the row is the address's high byte and the column can never carry into it, which is why the screen code can keep a row in `zp_screenline + 1` and nothing else. Port 0 reads and port 1 writes; the rest of the screen layer assumes port 0 is left selected with stride 1 ($\$11$ in `VERA_addr_bank`), so any routine that borrows port 1 must hand port 0 back that way.

The ordinary version 6 build is text only, on VERA's 80x60 screen, and it needed no window code of its own: `.s_scroll_vera` and the `scroll_window` opcode go through the same `.calc_window_rect` / `.sw_up_one` / `.sw_blank_row` helpers as every other target, and only the row copy (`.vera_copy_row`) is VERA's. One habit does not survive the window model, though. `print_line_from_buffer` used to print its first character through `s_printchar` and let the rest of the line ride on the address pointer that call left behind, because VERA auto-increments. Now that a window can scroll, `s_printchar` can end up inside `.s_scroll_vera`, whose copy loop leaves the pointer wherever it happened to finish — so every cell is addressed on its own, as on the other targets. Do not lean on VERA's auto-increment across a `jsr`.

The pictures screen

VERA is a tile machine, so the X16 draws pictures the way the MEGA65 does, and `pictures-x16.asm` is a port of the same engine (`Z6_PICTURES`; there is no `-fcm` to ask for, since VERA has no equivalent switch). The difference is that VERA has two layers, and Ozmoos uses both: the text stays on layer 1, exactly as in the text-only build, and the pictures live on layer 0 behind it. A tile is 16 pixels wide, 8 high and 4 bits a pixel, and layer 0's map is 64x32 of them.

The geometry is chosen to match the MEGA65's 80-column full colour screen, so that `dfrotz` at 80x25 and the MEGA65 build remain the references for both. VERA's scaling registers apply to the whole display rather than to one layer, so halving `VSCALE` (to 64) and cropping the display with `VSTOP` turns the 80x60 text mode into 80x25 of double-height rows on a 640x200 effective canvas. `SCREEN_HEIGHT` is therefore 25 under `Z6_PICTURES`, and the build reports interpreter number 6 (IBM), like the MEGA65's 80-column picture builds.

One picture tile is one logical 8x8 cell of the game's 320-wide art, with every pixel written twice as the tile is baked, so a picture keeps its intended size next to 8-pixel-wide text — the same doubling the MEGA65 does, but a whole cell at a time, because a VERA tile is 16 pixels wide where a full colour cell is 8. The tile store is the whole of VRAM's first 64 KB: 1024 tiles, with tile 0 reserved as the all-transparent tile an empty map cell shows, so a picture's run of tiles starts at 1. That is half the MEGA65's 2048, and it is enough: the worst live set any of the four games needs is about a thousand tiles (Arthur's frame plus a scene plus its status panel; the largest single picture, the sword title, is 751).

The colour arrangement is where VERA and the VIC-IV really part company. A layer 0 map entry is the tile's index (10 bits) and a palette offset (4 bits, plus the flip bits, which Ozmoos does not use), and the offset selects one of 16 banks of 16 colours in VERA's palette RAM. So a picture's palette bank rides in the map cells rather than in the pixels: the tiles hold the picture's own colour indices untouched (0 transparent, 1 to 15 straight into the bank), and nothing has to be rebaked when a bank is reassigned. Bank 0 is the text colours; a drawn picture takes one of banks 1 to 14, bumped alongside its tile run and reused alongside it, exactly as on the MEGA65.

Because the text sits on the layer in *front* of the pictures, drawing a picture has to blank the text cells it covers, or a window's painted background would hide the picture completely. It writes a space with the background nybble cleared, which VERA renders as transparent, so the picture behind shows through; `erase_picture` and `erase_window` put opaque background back and clear the layer 0 cells underneath. The result is the MEGA65's semantics — a picture replaces the text in the cells it covers — reached by different means. A fully transparent cell (`$ffff` in the cell map) is left alone on both layers, so a frame with a hole still shows the scene behind it.

Compositing a partly transparent cell over another picture works as it does on

the MEGA65, with one wrinkle. There, the store holds absolute palette indices and every bank is loaded at once, so a composite tile can freely mix two pictures' colours. Here a map entry carries a single palette offset, so a composite cell can only speak one bank's language: the pixels kept from the picture behind are translated into the overlay's bank, by exact colour match where the bank has the same colour and by nearest colour (the smallest sum of absolute red, green and blue differences) where it does not. The interpreter keeps a copy of every loaded bank in ordinary RAM for that comparison, so it never has to read the palette back out of VERA.

On quit the composer is put back — VSCALE, VSTOP, and layer 0 switched off — because the X16 build returns to BASIC rather than resetting the machine, and BASIC would otherwise come up on 25 double-height rows over whatever picture was last drawn.

Layer 0 moves with the text. `.s_scroll_vera` calls `pic_scroll_win_up` after it scrolls a window, so a drop-cap initial travels up with its paragraph instead of staying put and smearing its colours across the scrolled text; `erase_line` clears the layer 0 cells on the cursor's row (`pic_erase_line_cells`) as `erase_window` and `erase_picture` already did, so a picture behind erased text goes with it. Both touch whole map cells only and only VERA's write port, so the screen layer's port 0 is left as it found it.

A picture cell is two text columns wide, and a game may place a picture at an odd text column — Arthur centres its scenes. VERA's tile map is a 10-bit index, so the MEGA65's trick of storing each half-cell as its own 8-pixel tile would need more than the 1024 tiles the map can address. Instead an odd placement is drawn with **boundary tiles**: `.pic_fill_cells_odd` spans one more map cell than the picture is wide, and bakes each as the right half of one art cell beside the left half of the next (`.pic_bake_boundary_buf`), from the picture's own stored tiles. A picture whose copied run plus its boundary tiles would overflow the store falls back to even placement — an eight-pixel shift, invisible on a full-screen frame. Erasing an odd picture clears only the interior cells and leaves the two shared edge cells, each of which holds half the picture beside half of whatever it was composited over (a frame border), so the border survives a room change and the next scene re-composites its half over it.

One more thing separates the two layers. On the MEGA65 a picture and the text share one plane, so a transparent picture pixel shows the screen background (\$d021) directly. On the X16 the picture is on layer 0 and its transparent pixels (index 0) reveal VERA's backdrop — palette entry 0, which is black and is shared with text colour 0, so it cannot simply be recoloured. Left alone, Zork Zero's in-text insets (drop-cap initials, the banquet-hall statue, room icons) sit in a black box inside the white text window, where the MEGA65 and sfrotz show them on white. So a picture drawn over nothing has its transparent pixels filled with the screen background instead: `x16_screen_bg` follows the last `set_colour` background the way \$d021 would (`set_window` leaves it be, because a game draws an inset into a throwaway window whose own background is the default

while the text around it is white). `.pic_compute_bg_index` finds that colour in the picture’s palette bank; the insets have no white, so it usually is not there, and the colour is then injected into a free index the picture’s pixels do not use (`pic_used`, recorded while the tiles are copied) and the transparent pixels point at it. Only a direct picture’s own bank is written to; an adaptive picture shares the direct one’s bank and is left alone. Composited cells, a black screen background and fully-opaque pictures are all unchanged.

Pictures

Pictures are drawn on the MEGA65, with `Z6_FCM_MODE`, and on the X16 (see above). On the C64 and the Plus/4, `draw_picture` writes a “pic:N” note where the picture belongs, straight to the screen so that it never reaches the transcript, `erase_picture` paints the note out, and `picture_data` reports that no picture exists.

Everything in this section is the MEGA65’s arrangement unless it says otherwise; the X16 shares the design — the picture format, the index, the tile store, the allocator, the compaction and the adaptive palettes — and differs in the hardware it draws with and in how the pictures reach memory (the MEGA65 decrunches an archive of them into Attic RAM at boot, the X16 reads one uncompressed file from SD as each is drawn), both described in “The VERA screen on the X16” above and summarised at the end of this section.

The pictures come from the game’s blorb, converted by `tools/pics2asm.py`. Each PNG becomes a fixed-format picture (numbered up to 999; the format is described below), and the pictures of a disk are page-padded — so every one starts on an Attic page boundary — concatenated, and crunched into a single Exomizer archive, `pics<N>.bin`. `make.rb`’s `-pics` switch takes a blorb file (or, for the test pictures, a directory of numbered PNGs), runs the script, and sets `Z6_PICTURES`. Only an index is assembled into the interpreter — the picture numbers, the Rect placeholder sizes, the adaptive-palette flags, each picture’s size in Attic pages (`pic_pages`) and which disk its archive is on — so a set of a hundred and forty pictures costs it almost nothing.

One archive per disk compresses far better than the old scheme of one RLE file per picture, because Exomizer sees the redundancy *between* pictures — shared palettes, repeated tiles and cell-map runs — that a per-picture coder cannot. The whole set typically halves, and Zork Zero’s 396 tiny icons shrink more than threefold. That is what lets every shipped game fit one picture disk: Zork Zero and Journey needed two before (Zork Zero because 396 files overflowed a d81’s directory, Journey because its bytes overflowed one disk), and both now fit one.

The archives do not fit on the boot disk beside the story, the interpreter and the sound, so they get their own disk(s). `pics2asm.py` packs the pictures across as many `mega65_<game>pics_N.d81` disks as their archives need, capped by

block count, and `make.rb`'s `build_picture_disks` builds one d81 each, one archive on it. (Because a disk now holds one archive rather than one file per picture, the d81 directory limit that used to split Zork Zero no longer bites.) At boot, `pic_load_all` sweeps the disks in turn: for each it reads the archive into an Attic staging buffer at `$08510000`, then decrunches it forward onto the running Attic write pointer, under a "loading graphics" label with a slash progress bar. The bar ticks once every so many pages staged from disk — the slow part on a real 1581 — scaled through the assembled-in `picture_crunched_pages` so it is about thirty slashes wide whatever the set's size. For each disk `pic_load_all` tries the second drive (`boot_device + 1`) then the boot drive, reading the drive error channel to tell a present file from a missing one — a missing file `OPENs` "successfully" but reads back a stale buffer page, so the status byte lies — and only asks for a swap if neither drive holds it. `make.rb` mounts the first picture disk in drive 9 when it launches `xemu`, so a one-disk set loads with no prompt. The picture count and the picture numbers are both 16-bit: `.pic_index` is a word and the parallel `pic*` tables are indexed through `.pic_addr`.

The decruncher is a port of Exomizer's reference decoder (`exodec.c`) for the -P0 stream format, so `pics2asm.py` crunches with `exomizer raw -C -P0 -c -m 4096` to match. It reads the archive *forward* from the staging buffer through the 45GS02's [zp],z addressing and writes the plaintext forward into the picture area; a back-reference is read straight out of the plaintext already written (`src = output - offset`), so there is no ring buffer to maintain. `-m 4096` bounds a match to 4 KB back, which costs almost nothing here — page-padded, tile-deduplicated pictures match locally — and keeps the offset arithmetic to two bytes. The plaintext it produces is the padded concatenation the archive was built from, so each picture lands on exactly the page `pic_pages` predicts.

A restart does not pay the load twice. `z_ins_restart` reboots the machine and reloads the interpreter from disk, so `pic_load_all` runs again on every "restart" command and every death — but Attic RAM survives the reset, and the pictures are still sitting in it. This time the page tables that say where each picture landed need no rescue: `pic_page_lo` and `pic_page_hi` are computed at every boot by summing the assembled-in `pic_pages` (each picture's size in pages, known at build time because the *uncompressed* size is) up from `PIC_ATTIC_PAGE`, so they are correct whether the pictures were just decrunched or have been sitting in Attic since a previous boot. All that has to survive the reboot, then, is the answer to "are the pictures still there?", and `$08300000` holds it: an eight-byte signature, with the pictures starting one page above at `PIC_ATTIC_PAGE`. On boot a matching signature skips the disks entirely, swaps and all. This is `reu_filled`'s idiom from `disk.asm`, a signature that outlives the reboot to say the data is already there, with two differences: it cannot use `game_id`, which is inside `!ifdef VMEM` and so does not exist on the MEGA65, and uses the story's serial and checksum instead — they identify the story file just as well, and Ozmo never rewrites either, unlike `flags_2` or the screen dimensions; and the signature is written *last*, after the pictures, so a load interrupted partway leaves none claiming they are good. The check runs

on every boot rather than only after a restart, so a cold reboot of the same game skips the load as well. The X16 needs none of this, having no preload at all.

A blorb holds more than PNGs. Its Rect resources are placeholders with no image, only a width and height; a game reads those sizes with `picture_data` to lay real pictures out, and Arthur's whole room frame is built that way, so `pics2asm.py` emits a table of them (in cells, $\text{ceil}(\text{pixels}/8)$) that `picture_data` answers from and `draw_picture` treats as invisible. Its APal chunk lists the adaptive-palette pictures (see below). Reading the blorb directly rather than a hand-extracted PNG directory is what makes both of these available.

A picture holds the cell dimensions, the number of unique tiles, a 48-byte palette, a cell map of two bytes a cell, and then the tiles. Identical cells are stored once. Each picture is padded up to a page boundary; the pictures of a disk are concatenated in that padded form and the whole run is Exomizer-crunched into the disk's archive, so the compression works across pictures, not just within one. Once decrunched into Attic RAM the layout is exactly this padded concatenation, and the padding is why each picture begins on a page whose number the index already knows.

Two things about the pixels:

- A pixel is four bits, so two pixels share a byte on disk and in Attic RAM, while in the tile store a pixel is a whole byte. On the 80-column screen each logical 8x8 cell of the source becomes two screen cells — its left half and its right — and the halves are stored undoubled: a tile on disk is then 4 source pixels a row, 16 bytes, and `.pic_copy_tiles` writes every pixel twice as it bakes the 64-byte store tile, doubling the picture to its 320-wide visual scale for free. (Under `-fcm:40` a tile is the whole 8x8 cell, 32 bytes, written once.) Deduplication happens on the stored tiles either way. Storing four bits a pixel halves the picture set before it is compressed, and Arthur's does not otherwise fit on a d81.
- A colour index of 0 is transparent, and 255 is taken from colour RAM, so neither may be a real colour. A picture therefore has at most 15 colours. The indices are the PNG's own, not compacted: a pixel is its palette index straight, and the 48-byte palette is stored in that same index order. Keeping the indices is what lets an adaptive picture borrow another's palette (below). Arthur's pictures put transparency at index 0 and never colour index 0.

Drawing a picture reads its header and palette straight out of Attic RAM, gives the window a run of the tile store, copies the tiles down into it, and fills the window's cells. In Full Colour Mode a cell's screen code is its tile's address divided by 64, so a code is `FCM_TILE_CODE_HI`'s worth of high byte plus the tile's index: the store's base is a multiple of \$4000 either way, so nothing carries.

A cell holds exactly one tile, so a picture drawn over another replaces those cells outright. A cell that is transparent through and through is the exception: `pics2asm.py` writes it into the cell map as `$ffff`, and `pic_fill_cells` leaves such a cell untouched, so a frame with a hole drawn over a scene shows the scene through the hole. That is how Arthur composites — it draws the scene, then draws the frame, whose middle is a transparent window onto the scene behind. A partly transparent cell is still one opaque tile, and its transparent pixels show the screen background rather than the picture behind. Drawing is clipped to the screen: `pic_fill_cells` and `pic_erase` stop at the last row and column, so a picture placed partly, or from a bad coordinate wholly, off screen cannot scribble past screen RAM into the interpreter.

`picture_data` is what makes a picture land in the right place. It reports a picture's height and width, which the game halves against the window's size to centre it. The standard calls those two words pixels, but a pixel here is whatever unit the rest of the screen model counts in, and Ozmo counts characters: the header advertises an 80x25 screen (40x25 under `-fcm:40`) and `get_wind_prop` answers 1x1 for the font size, so the size must come back in cells. On the 80-column screen the blob header and the Rect tables already carry the doubled widths, so the game lays everything out in the 80-wide grid without any conversion.

Each drawn picture gets its own run of the tile store and its own palette bank, allocated together and reused together. The bank is `16 * bank`, bank running 1 to 14 (bank 15 is skipped, because its top colour would be pixel value 255, which Full Colour Mode takes from colour RAM). A per-picture bank, rather than the one-bank-per-window arrangement it replaced, is needed because a window holds several pictures at once — Arthur's frame and two side bars all live in window 7 — and a shared bank had them overwrite each other's colours. A picture's pixels are its palette indices straight, so the tile copy adds the bank's base to each non-zero pixel as it goes, leaving a 0 alone because it is transparent. The DMA engine can neither add nor expand a nybble, so that copy is done by the CPU; at 40 MHz even a full screen picture is a blink.

The bank travels with the tile run: the allocator reuses a window's run, and its bank, only when the same picture is redrawn into it; a different picture takes a fresh run and bank, bumped from where the last one ended. The store holds 2048 tiles in a `Z6_PICTURES` build, which is what Arthur's interface needs: it keeps a border, a scene and a status panel on the screen together, and they came to 1063 tiles the first time all three were live. With the old store of 1024 the allocator wrapped and the border's tiles landed on the scene's. On the 80-column screen the same live set measures 1894 tiles — splitting every logical cell in two does not double the unique count, because the halves deduplicate — and `pics2asm.py`'s `-stats` switch prints the per-picture numbers without writing anything.

Even 2048 tiles run out, though, because a game may draw one picture over another and keep both: Arthur draws the Merlin scene centred inside the sword

picture, whose gold frame stays visible around it, and the two together are more than the store holds. So when a fresh run will not fit, the allocator compacts the store rather than wrapping over it (`.pic_gc`). It marks every tile that a cell *outside* the incoming picture's rectangle still shows, moves those survivors down to the bottom of the store in ascending order — a survivor can only move down, so the copy is safe in place — and repaints the cells that show them; the new run then lands above the survivors. Cells the new picture is about to cover are not marked, since it is about to overwrite them anyway, and if their tiles are shared with cells outside (Arthur's borders repeat), the sharing marks them survivors in any case. The sweep works from a bitmap of live tiles, a per-byte prefix count of the survivors below it and a popcount table. Only if the survivors plus the new picture still do not fit does the old wrap-to-zero happen, which can then only spoil a picture that is no longer the newest on screen. The palette allocator still simply wraps, with the same consequence.

Not every tile a draw consumes comes out of that run, and the difference is worth spelling out, because it cost a long hunt. The allocator reserves a picture's own tiles and nothing more, but both draw paths then bake further tiles from `pic_next_tile` as they go — a composite where a partly transparent cell falls over an existing picture, and on the X16 a boundary tile per cell of an odd-column placement. Those belong to no window's run, so the store can genuinely run out in the middle of a draw, where there is no opportunity to compact: the picture's own run is already placed and half its cells are written. The X16's baker therefore *saturates* rather than wrapping (`.pcf_alloc_and_write`): `pic_next_tile` stops at `PIC_MAX_TILES` and the bake reports failure, and the caller degrades for the cells it could not bake — the even paths keep their own un-composited tile, the odd path keeps whatever was behind. That is self-correcting, because a saturated `pic_next_tile` fails the allocator's fit check, so the next picture drawn compacts the store and the bakes have room again; the cost is cosmetic, confined to a nearly full store, and the alternative was corruption.

It used to wrap to the bottom of the store, and that destroyed Arthur's frame a few rooms into a game. Everything below is live — the picture's own run, and the frame, which is drawn once at boot and never redrawn and so sits at the very bottom — so the wrap landed squarely on it. Two properties made it hard to see. The damage surfaced two rooms *after* the wrap, only once the allocator had marched back up into the frame's tiles, which pointed suspicion at the wrong event. And the wrap disabled the compactor that would have saved it: it left `pic_next_tile` low, so the fit check always passed and `.pic_gc` never ran again. The lesson generalises past this engine — an allocator that wraps onto live data is a bug rather than a fallback, and a wrap that erases the “store is full” signal takes the recovery path down with it. The MEGA65's baker still wraps: it has 2048 tiles and half the X16's per-picture cost, so it has not bitten, but the defect is the same one and the fix to port is the X16's.

Some pictures do not carry their own colours. The blorb's APal chunk lists

the “adaptive” pictures — Arthur’s frame and side bars — which are drawn in the palette of the last direct (non-adaptive) picture instead of their own. That is how one UI element recolours to match whichever scene it borders; in the blorb the frame’s own palette is a placeholder of plain primaries. The interpreter remembers the last direct picture’s bank (`pic_direct_base`), and for an adaptive picture it skips loading a palette and bakes the tiles into that bank. Because the indices are never compacted, the frame’s index 5 means the same colour as the scene’s index 5, and the borrowed palette lands right.

What the X16 does differently

The X16 shares all of the above: the same converter, the same index, the same allocator, compaction and adaptive palettes, and the same clipping. It differs in four ways, three of them consequences of the hardware and one of the machine it runs on.

- **The tiles and the cells.** A tile is a 16x8 VERA tile rather than a 64-byte full colour cell, so a logical art cell is one tile, not two, and `pics2asm.py`’s `-x16` mode writes one cell-map entry and one 32-byte tile per cell (the pixels undoubled; the interpreter doubles each as it bakes). The palette is 32 bytes in VERA’s own format instead of 48 in the VIC-IV’s. The store holds 1024 tiles, with tile 0 reserved as the transparent tile.
- **The palette bank is in the map entry, not in the pixels.** Which means a composite cell can carry only one bank, and the pixels taken from the picture behind have to be colour-matched into the overlay’s — see “The VERA screen on the X16”.
- **The pictures are loaded when they are drawn, not at boot.** There is nowhere on the X16 to preload half a megabyte of pictures: the story already fills most of the 512 KB of banked RAM. But the pictures are on an SD card, not a floppy, so there is no need to. Each picture is an ordinary uncompressed file in the game’s directory, and the engine reads it into the staging banks the first time the game draws it, remembering which picture is staged so that redrawing the frame or a status icon never touches the card again. The files are therefore not compressed and there is no decruncher, no picture archive, no picture disk, and no disk swapping — Zork Zero’s 396 pictures simply sit in the directory. The cost is that `picture_data` cannot read a size out of the staged picture, because it is usually asked about a picture that is not the staged one, so `pics2asm.py` assembles the sizes in as `pic_width` and `pic_height` tables (in text cells, as the MEGA65 reports them).
- **The text is a layer in front, so drawing must blank it,** and erasing must put it back. On the MEGA65 the text and the picture share one plane, and a tile simply replaces a character.

Mouse

The mouse (z-spec 10.3) is drawn on the two screens that can show a pointer: the MEGA65's full colour screen and the X16's pictures screen. Both are in `asm/mouse.asm`, guarded by `Z6_MOUSE`, which is set for `Z6_FCM_MODE` or for the X16 with `Z6_PICTURES`; text-only version 6 has no pointer on either machine. A game asks for a mouse by setting bit 5 of Flags 2, and the interpreter leaves that bit set only where it can honour it. `z_init` reads the bit and, if the game wants a mouse, calls `mouse_enable`, which shows the pointer and sets `mouse_active`; a game that does not ask gets no pointer and its clicks are ignored. (Arthur, Zork Zero and `testz6` — whose `read_mouse` test asks — all get one.)

Everything the standard asks for is the same on both machines and is written once: the press edge that turns a held button into a single click, the cell the click landed in, the words written into the header extension table, and the window the mouse is confined to. Only reading the hardware and moving the pointer differ, and that is one routine, `mouse_read`, which `mouse_poll` calls and which returns the button mask with the pointer's pixel position updated.

The MEGA65's mouse

The pointer is the 1351 or Amiga mouse on control port 2. The MEGA65 exposes its paddle lines directly at `$d620` and `$d621`, with no SID or CIA multiplexing to arrange, and the button is the port 2 fire line, `$dc00` bit 4. Port 2 is the clean joystick port; port 1 shares its lines with the keyboard matrix, so its fire bit `$dc01` bit 4 reads keyboard row 4 as well and cannot tell a click from a SPACE. The paddle value is a six bit counter that wraps, so `mouse_poll` cannot read an absolute position; it remembers the previous reading, folds the difference into a signed step of -32 to 31, and accumulates a pixel position clamped to the screen: 640x200, or 320x200 under `-fcm:40`. Sprite X coordinates stay in the 320-wide space under H640 — one sprite unit covers two physical pixels — so `mouse_place_sprite` halves the position on the 80-column screen and the pointer moves in two-pixel steps.

The pointer is sprite 0. Its bitmap is a small red arrow (red so it stays visible over the white status line), and it is addressed through the VIC-IV's 16-bit sprite pointer (`$d06c` to `$d06e`, with the `SPRPTR16` bit set), because the sprite pointer list normally sits at the top of screen RAM, which on the full colour screen is picture and text data. There is no hardware that makes a sprite follow the mouse; the MEGA65 KERNAL moves it in an interrupt and so does Ozmo, from `mouse_poll`. On quit the sprite is switched off along with the full colour mode bits, so no arrow is left on the BASIC screen.

The X16's mouse

On the X16 the kernal owns the mouse, and `mouse_read` is a single call into it. `mouse_config` (\$ff68) shows the pointer and gives it its bounds, taken in units of 8 pixels: 80 by 25, which is the 640x200 pictures screen. `mouse_get` (\$ff6b) fills four zero page bytes with the position and returns the button mask in the accumulator; the four bytes are the kernal's own API registers r0 and r1 at \$02, which its interrupt handler saves and restores around its use of them, so they are safe to borrow. The kernal's default interrupt handler also calls `mouse_scan` and moves the pointer itself, and Ozmoos never replaces the interrupt vector on the X16, so there is no sprite for Ozmoos to place and no paddle counter to difference.

The pointer is VERA sprite 0, and the kernal keeps its image at video RAM \$13000, which is clear of both the tile store and the layer 0 map (see the X16 memory map above). `mouse_enable` therefore only has to switch the sprite layer on in `DC_VIDEO`. One detail is worth knowing: because the screen is fewer than 31 cells high, the kernal doubles its internal y coordinate and `mouse_get` halves it back again. That is exactly right for a screen whose `VSCALE` is halved — the pointer moves at the same physical speed as on any other X16 screen, and reports a position of 0 to 199. On quit the pointer is hidden with `mouse_config` again, which also puts the system management controller back to sending key codes only.

Clicks, and how they reach the game

`mouse_poll` is called from `getchar`, the routine every input path waits on, so the pointer follows the mouse through `read`, `read_char` and the [MORE] prompt, and a button press is delivered as input character 254 the moment it happens. On a click the cell the pointer is over is written into words 1 and 2 of the header extension table, as the standard requires; `read_mouse` reports the same position and the button, and `mouse_window` confines later clicks to one window. A click ends a line only if 254 is a terminating character, which the game arranges through the terminating characters table – see below.

The terminating characters table (header \$2e, z-spec 10.5.2.1) is a list of the function key codes that end an input line, in addition to Return. The valid codes are 129 to 154 and 252 to 254, and the special value 255 means “any of them”. `parse_terminating_characters` had rejected everything above 140 and, for the 255 wildcard, activated only the ordinary function keys, never the mouse clicks 252 to 254 – so a click could not end a line, whichever way a game asked. It now accepts the whole valid range, and the default set it turns on for the wildcard includes the mouse clicks on a screen that has a mouse, so both a game that lists the clicks (Arthur) and one that says “any function key” (Zork Zero) work.

The stack

The exact layout of the stack is an implementation detail, not part of the Z-machine specification. However, the Z-machine specification does say a few things about how the stack should *work*. Each routine call creates a stack frame. Instructions can push any number of values onto the stack, and pull values from the stack, or read or change the top value without pushing or pulling it. A value is always word-sized. Pushed values can only be accessed by instructions in the same routine (= same stack frame). When a routine returns, all values which have been pushed inside that routine and are still on the stack are discarded. The stack resides outside the Z-machine, and there is no way for a Z-machine program to check if there are values on the stack, if there is room for more values, how many stack frames there are etc.

The stack in Ozmoos consists of a number of words. The default size is 512 words. With each procedure call that is made, a new stack frame is created. The first stack frame begins at the first address of the stack (label `stack_start`).

A stack frame consists of the following parts:

1. The current values of all local variables of the routine (0-15 words).
2. Frame header (2 words)

Frame header A has the following parts:

Byte 0, bit 7: 1 = The return value of this procedure call should be written into a variable

Byte 0, bit 4-6: In version 5+, the number of parameters this procedure was called with (0-7)

Byte 0: bit 0-3: The number of local variables this procedure has (0-15)

Byte 1, bit 7: 1 = this routine call was done due to an input interrupt

Byte 1, bit 0-2: Bit 16-18 of the return address

Frame header B:

Bit 0-15 of the return address (highbyte first).

3. Pushed words (0 or more words)
4. Number of pushed bytes (1 word)

The first frame lacks part 1 and 2. The current frame lacks part 4.

Apart from this, Ozmoos also has a few variables on zeropage to keep track of the stack:

`stack_ptr` (word): a pointer to where part 3 of the current stack frame begins

`stack_pushed_bytes` (word): A counter of how many words part 3 of the current frame consists of.

`stack_has_top_value` (byte): If the value is 1, the last value pushed onto the stack in the current frame hasn't been put on the actual stack, but resides in the variable `stack_top_value` instead.

`stack_top_value` (word): If `stack_has_top_value` is 1, this contains the top value on the stack.

The `stack_top_value` variable has been added for performance reasons.

Frame 0 isn't a full stack frame. When a Z-machine interpreter begins execution of a program, it just starts at the start address given in the header of the story file, with an empty stack. The start address can be in the middle of a routine. This means the interpreter doesn't know of any local variables, and there is no return address - it's not legal to perform a return from this routine. This "fake" routine is just a simple way to get started, and the only thing you want to do in this routine is call a "real" routine and then perform quit. Inform 6 adds such a routine automatically, and it calls the routine "main" which must be defined in the Inform 6 program and then quits. To make things more confusing, this little "fake" routine is called "main" when disassembling the Z-code program.

Let's go through an example program and what it does to the stack. We run it until it performs `@read_char` and then we pause and look at the stack. This is a version 5 program.

Inform's built-in "fake" main routine (on address \$4ec) calls our main routine (on address \$514). This causes Omzoo to close the current frame (Frame 0, which isn't a full stack frame) by moving the top value from `stack_top_value` (if any) to the stack, and writing the total number of pushed bytes as a word to the stack. In this case, there are no values on the stack, and no value in

```

!% ~~$
!% $OMIT_UNUSED_ROUTINES=1

[foo x y;
  @push 9;
  @push 3;
  print x;
  new_line;
  print "Press a key: ";
  @read_char y;
  print y;
];

[main;
  @push 7;
  foo(5);
];

```

Figure 2: stackdemo.inf, Inform 6 source code

```

Main routine 4ec, 0 locals

4ed: e0 3f 01 45 ff      CALL_VS      514 -> Gef
4f2: ba                  QUIT

Routine 4f4, 2 locals

4f5: e8 7f 09            PUSH        #09
4f8: e8 7f 03            PUSH        #03
4fb: e6 bf 01            PRINT_NUM   L00
4fe: bb                  NEW_LINE
4ff: b2 ...              PRINT        "Press a key: "
50a: f6 ff 02            READ_CHAR   -> L01
50d: e6 bf 02            PRINT_NUM   L01
510: b0                  RTRUE

Routine 514, 0 locals

515: e8 7f 07            PUSH        #07
518: da 1f 01 3d 05      CALL_2N     4f4 (#05)
51d: b0                  RTRUE

```

Figure 3: stackdemo.zasm, disassembly

Stack contents:			
3a00:	00 00	Number of bytes pushed	Frame 0
3a02:	80 00	Frame header A	
3a04:	04 f1	Frame header B	
3a06:	00 07	Pushed word 0	Frame 1
3a08:	00 02	Number of pushed bytes	
3a0a:	00 05	Local variable 0	
3a0c:	00 00	Local variable 1	
3a0e:	12 00	Frame header A	Frame 2
3a10:	05 1d	Frame header B	
3a12:	00 09	Pushed word 0	
stack_ptr: 12 3a			
stack_pushed_bytes: 00 02			
stack_has_top_value: 01			
stack_top_value: 00 03			

Figure 4: stackdemo, stack contents

stack_top_value, so Ozmoos just copies the value in stack_pushed_bytes (0) to the stack.

Ozmoo then starts a new stack frame - frame 1. The main routine at \$514 has 0 local variables, so it doesn't reserve any space for part 1. It then writes the frame header to the stack. Frame header A is \$80, \$00 because the procedure call is made using call_vs which expects the return value to be written into a variable, there are no arguments and the routine has no local variables, the routine was not called due to an interrupt, and bit 16-18 in the return address are all 0. Frame header B holds the value \$04f1, which points to the last byte of the call_vs statement, holding the variable number where the result is to be stored upon return.

Program execution now continues in our main routine at address \$515 (the routine's official start address is 514, but the first instruction is on address \$515.) The first instruction pushes the value 7 onto the stack. The second instructions performs a call_2n instruction to call the foo routine with one argument, which is the value 5. We can see that the value 7 has been pushed onto the stack, and that the counter says a total of 2 bytes have been pushed onto the stack in this frame.

Ozmoo now opens stack frame 2. The routine being called (the foo routine located at address \$4f4) has two local variables, so Ozmoos makes room for two words on the stack for part 1 of the frame. The first variable gets the value 5, since the procedure was called with this as an argument. Ozmoos then adds part 2, the header. Frame header A gets the value \$12, \$00 because the routine was not called with an instruction which expects the return value to be written into a variable, the routine call was made with 1 parameter, the routine has 2 local variables, the routine was not called due to an interrupt, and bit 16-18 of the

return address are 0. Frame header B holds the value \$051d, which is where execution should resume upon return from this routine.

The foo routine pushes the value 9 and then the value 3. At the time when we inspect the stack, the value 3 is held in `stack_top_value`. If we were to push another value, or call a procedure, this value would be written to the stack as well.

There are two more `print` instructions and a `new_line` instruction before `read_char`, but none of them interfere with the stack. When we hit `read_char`, we paused the program and inspected the stack.

Virtual Memory

This chapter is based on a document written by Steve Flintham.

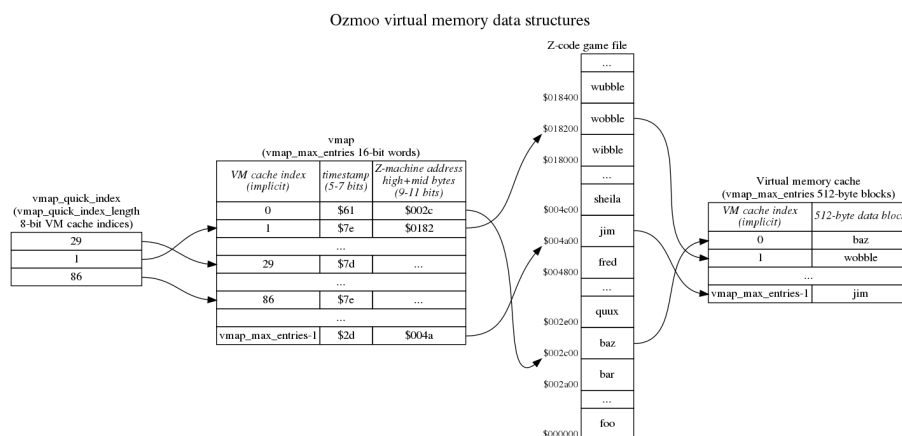


Figure 5: Virtual memory overview

The basics

The virtual memory subsystem is only used for the game's static and high memory, which are read-only. The game's dynamic memory is always held entirely in RAM.

The virtual memory code does most of its work in the `read_byte_at_z_address` subroutine. (This can be seen in `vmem.asm`; there are multiple versions of this subroutine in the file because conditional assembly is used to allow non-VM builds, but the VM version starts about halfway down.) It's entered with a 24-bit Z-machine address in A, X and Y (high byte in A, middle byte in X, low byte in Y) and returns the value at this address in A. Also, upon returning, Y is 0 and `mempointer` (a 2-byte zero page pointer) is set so that `"lda (mempointer),y"` returns the byte at the 24-bit address given.

Note that `read_byte_at_z_address` isn't called for every single read from Z-machine high and static workspace; there are subroutines layered on top of it which only call it when necessary.

Virtual memory is handled in 512-byte blocks, which are always aligned at a 512-byte boundary in the game file and in memory. This means that every VM block has a Z-machine address of the form `$abcd00`, and the least significant bit of `d` is always 0.

The VM system has a cache with `vmap_max_entries` 512-byte blocks of RAM to use to hold blocks read from disc. There's a parallel data structure called the `vmap` with `vmap_max_entries` 16-bit words in it to track what block of game data is in each cache block. At the most basic level, we can imagine that if block 4 of the cache contains the 512-byte block starting at `$018200` in the game file, `vmap[4]` contains `$0182`.

If we want to access the byte at `$018278`, we round that down to the previous 512-byte boundary to get `$018200` and then search through all the entries in `vmap` to see if any of them contain `$0182`. In this case, entry 4 does, so we'll set `mempointer` to `cache_start_address+4*512+$078`, load the value at that address into `A` and return. Where does the `$078` come from? This is the low 9 bits of the address we were given, which can be thought of as an offset to the byte of interest within this 512-byte block.

What if none of the `vmap` entries contains `$0182`? In that case we need to pick one to overwrite; let's say we pick entry 7. Because we only use virtual memory for read-only data, we can just read the 512-byte block at offset `$018200` in the game file into memory at `cache_start_address+7*512` (overwriting whatever was there), set `vmap[7]` to `$0182` and return with `A` set to the value held in address `cache_start_address+7*512+$078`.

At the most basic level, that's all there is to it. But in practice there are some additional details we need to take care of.

Timestamps

We need to be able to make a sensible decision about which cache block we're going to overwrite when we need to read a block of data in from disc because it's not in the cache. What we want to do is keep hold of cache blocks we've recently used, and instead discard a block we haven't used in a while, on the reasonable assumption that we're more likely to use a block in the near future if we've recently used it.

To implement this, each `vmap` entry also contains a timestamp. There's a global "current time", called `vmem_tick`, and every time we access a cached block its timestamp is set to `vmem_tick`. `vmem_tick` is incremented whenever we need to read data from disc because it's not in the cache. Note that several entries

in the vmap can therefore share the same timestamp - if we access cache blocks 4, 8 and 22 without needing to read anything from disc, all of those will have the same timestamp (the current value of `vmem_tick`).

Also, there's a mechanism to keep `vmem_ticks` from getting too big for the space available. For a z3 game, we have 8 bits for storing tick values for each vmem block. We start with the value 0 and we can just increase it until we reach 255. When we hit 256, we change the tick counter to 128, and we reduce the tick value in all vmem blocks by 128. Since we're using unsigned numbers here, a block which had the tick value 27 now gets 0. A block which had tick value 150 now gets 22 etc. The next block we read from disk gets the tick value 128, the next after that gets 129 etc. Next time we reach 256, we just repeat the above procedure.

With that infrastructure in place, when we need to overwrite a block in the cache with a new block from disc, we can pick a block with the oldest timestamp. (Not *the* block with the oldest timestamp, because as noted above several blocks can share the same timestamp.) There's one caveat, which is that if the Z-machine program counter is currently pointing into a cache block, it is exempt from being overwritten - it would obviously be a bad thing to overwrite the instructions currently being executed with some arbitrary data! I won't go into too much detail on this here, because it's probably best discussed in the context of the routines layered on top of `read_byte_at_z_address`.

The timestamps are packed into the high bits of the 16-bit entries in vmap, with the low bits representing the high and mid bytes of the Z-machine address. For Z3 games, where Z-machine addresses only have 17 bits (a maximum game size of 128K), only 9 bits of the vmap entry are needed for the high and mid bytes of the address and 7 bits are available for the timestamp. But note that the lowest bit of the address is always zero since the memory is divided into 512-byte blocks. Assuming this zero we can instead store the address in 8 bits and increase the time stamp resolution to 8 bits.

Larger versions of the Z-machine need more bits for the high and mid bytes so fewer bits are available for the timestamp; a Z8 game (with 19 bit addresses for a maximum game size of 512K) only has 6 bits available for the timestamp. This limited timestamp resolution is the reason why `vmem_tick` is only incremented when a block needs to be read from disc, not every time `read_byte_from_z_address` is called.

`vmem_tick` is actually held in memory "pre-shifted" for convenience of using it to compare or update the high byte of the 16-bit entries in vmap, so rather than always incrementing by 1 at a time, it increments by 1 at a time for Z3 games, 2 at a time for Z4 and Z5 games and 4 at a time Z8 games. (In the code, the increment is a constant called `vmem_tick_increment`.)

The quick index

In general we have to do a linear search of vmap in order to see if a particular 512-byte block of the game is already in RAM (and where in RAM it is, if it's in RAM). The vmap can contain up to 255 entries (depending on platform), and it's likely to contain at least 64 entries, representing 32K of cache, so this is potentially quite slow.

It's quite likely that there's a relatively small “working set” of 512-byte blocks which we're going to be accessing over and over again. For example, maybe one 512-byte block contains a Z-machine function with a loop in, and inside that loop we call another Z-machine function which lives in a separate 512-byte block.

We therefore maintain a “quick index” containing the cache indices of the blocks we've accessed most recently. (There are `vmap_quick_index_length` entries in this list, which in practice means there are 6 entries.) We look at the corresponding entries in vmap first to see if those entries have the 512-byte block we're interested in, and if they do we can avoid doing the full linear search. Whenever we have to do the full vmap search, we overwrite the oldest entry in the quick index with the index we found from the full search. The quick index is effectively a circular buffer of `vmap_quick_index_length` entries, with `vmap_next_quick_index` pointing to the oldest entry.

As a consequence of this, the vmap entries pointed to by the quick index are those with the most recent timestamps.

Efficient access to vmap

Each vmap entry is a 16-bit word, so the obvious way to store vmap would be:

```
low byte of entry 0
high byte of entry 0
low byte of entry 1
high byte of entry 1
...
```

The disadvantage of this is that when we're using index registers to step through vmap, we need to do double increment (or decrement) operations:

```
ldy #0
.loop
lda vmap,y ; get low byte of entry
lda vmap+1,y ; get high byte of entry
...
```

```

iny
iny
cpy #max_entries*2
bne loop

```

We don't actually need the two bytes to be adjacent in memory, and so instead we have two separate tables. `vmap_z_l` stores the low bytes:

```

low byte of entry 0
low byte of entry 1
...

```

and `vmap_z_h` stores the high bytes in the same way.

With this arrangement, we don't need double increments (or decrements) to step through the table:

```

ldy #0
.loop
lda vmap_z_l,y ; get low byte of entry
lda vmap_z_h,y ; get high byte of entry
...
iny
cpy #max_entries
bne loop

```

This also has the advantage that we can access up to 256 entries using our 8-bit index registers, instead of 128 entries if we had to do double increment/decrement. The Commodore 64 version of Ozmoos doesn't need this, but the Commodore 128 version does. Acorn sideways RAM version benefits from it and there are a couple of minor tweaks to the code to make this work. (If you know there are <128 entries, you can use `dex:bpl` to control looping over the table and avoid needing a `cpx #255` to detect wrapping.)

Banking

On the Commodore 64 and Commodore 128 there's an additional wrinkle because some blocks of RAM are hidden behind the I/O area and the kernel ROM and there's a mechanism to copy those blocks of RAM into the so-called vmem cache in always-visible RAM when necessary.

The `first_banked_memory_page` variable stores the high byte of the first 512 block of RAM that isn't always visible. On the C64 this is `$d0`, since the

I/O registers are located from \$d000, and unrestricted reading/writing to these memory position cause all kinds of trouble.

When `read_byte_at_z` address detects that we want to read data from a page in the non-accessible memory (\$d000-\$ffff) then we will copy that page (256 bytes) from the non-accessible memory to one of the pages in the vmem buffer. This is done by `copy_page` (in `memory.asm`), which can copy a page securely from and to any memory position using memory banking as needed.

Continuous virtual memory optimization

Copying pages to the vmem buffer is rather expensive, so if possible, we would like the pages that are used the most to be in regular, unbanked RAM, where no such copying is necessary to access the data.

Ozmoo has a feature called continuous virtual memory optimization, which can help with this. It keeps statistics on how many times each block in virtual memory has been requested. After 256 block requests have been made, it swaps the most used block in banked RAM with the least used block in unbanked RAM. Then it resets the statistics, and repeats the cycle. After about 20 such swaps (45 on C128), it slows down the pace and starts waiting for 768 block requests before making a swap.

This ensures that the central parts of the game are gradually moved to unbanked RAM within the first 10-20 moves or so. After that, the system will still be able to adapt to other parts of the game file becoming more important, but it happens at a slower pace.

REU Boost

When Ozmoo is running on a Commodore 64 or 128 with an REU to cache game data, the virtual memory model is no longer optimal for the task - copying data from the REU is so fast, that it's not worth the trouble of tracking when each block was last accessed etc.

With the REU Boost feature, we include a simpler virtual memory model. Essentially, it's a circular buffer, holding a few KB of data. If a page we need can't be found in the buffer, we replace the page that has been in the buffer the longest with the page we need.

Ozmoo decides when the game starts, if game data will be cached to an REU. If so, and the interpreter has been built with the REU Boost feature, it switches to the REU Boost virtual memory model.

Sound

While several Infocom games would play high and low-pitched beeps, a few games had extended sound support using sample playback. Ozmoo supports the basic sound effects (beeps) on all platforms, and extended sounds on the MEGA65.

Sound support is implemented in `sound.asm`. Extended sounds also use `sound-wav.asm` to parse sample files stored in the WAV format. The WAV files need to be 8 bit, mono. Audacity can be used to export wav files in the correct format:

- Select “File/Export/Export as WAV” from the main menu
- Select “Other compressed files” as the file type
- Select “Unsigned 8-bit PCM” as the encoding
- Save the file

If extended sound is to be used, then `make.rb` must be called with the `-asw path` switch. If set, then all `.wav` files in the `path` will be added to the `.d81` floppy created for the MEGA65, and the `SOUND` assembly flag will be set when building Ozmoo.

For legacy reason AIFF is also supported, using `sound-aiff.asm`, and the `-asa path` switch. Code has also been added to detect and handle sound effects in *The Lurking Horror* game from Infocom, which isn’t standard compliant.

When compiled with extended sound support, Ozmoo will preload all sound files during startup into the MEGA65’s attic memory, and then copy each sound effect into fast memory on demand when the `@sound_effect` command is used in the game code.

The effect being played must live in fast RAM, at the foot of a bank: the audio DMA’s stop address is only 16 bits wide, so a sample cannot straddle a bank boundary and cannot be played from Attic RAM at all. Which bank that is depends on the build, and is the constant `SOUND_FASTRAM_BANK`. It is bank 4 everywhere except a version 6 build with pictures, where the tile store needs banks 4 and 5 and the sound effect moves down into bank 1.

Smooth Scrolling

This feature adds smooth scrolling support for the c64 target. When active, text is scrolled up one pixel (raster line) per frame rather than an entire character (text row) at a time, providing a “smooth” visual experience. The user can toggle whether smooth scrolling is active using the F2 key during the game.

Smooth scrolling is activated at program startup if the support was included. The code for the feature is in the new file `smoothscroll.asm`. Moving the screen data is performed using the existing scrolling code in `screenkernal.asm`, while the fine-scrolling manipulation is performed in a raster interrupt. This means that some program processing, including printing to the screen, could occur while scrolling is being performed. This helps keep an overall smooth effect when multiple lines are being printed and scrolled (such as with a long room description), but it also means that some coordination is necessary to prevent visual glitches when:

- the program wants to scroll again while scrolling is in progress
- the program will disable interrupts
- the program will perform REU access, which pauses the CPU

One mechanism provided to handle this is the `wait_smoothscroll` function, which can be called to wait until any current scrolling activity has completed. Another is the `smoothscroll_off` and `smoothscroll_on` functions, which can be used to temporarily disable and later re-enable smooth scrolling. The latter are used in the macros `disable_interrupts` and `enable_interrupts`, which are typically used in place of explicit `sei` and `cli` instructions.

The inclusion of smooth scrolling adds about 460 bytes to the interpreter size.

Accented Characters

Ozmoo has some support for using accented characters in games. Since the Commodore 64 doesn't really support accented characters, some tricks are needed to make this work.

The Z-machine, which we are emulating, uses ZSCII (an extended version of ASCII) to encode characters.

The Commodore 64 uses PETSCII (a modified version of ASCII) to encode characters. PETSCII has 256 different codes (actually less, since some code ranges are just ghosts, i.e. copies of other code ranges). Some PETSCII codes are control codes, like the ones to change the text colour or clear the screen. 128 of the PETSCII codes (plus some ghost codes) map to printable characters. A character set contains these, plus reverse-video versions of all 128, adding up to 256 characters all in all.

So, all in all there are basically 128 different characters, and that's it. They include letters A-Z, digits, special characters like !#\$%& etc, some graphic characters, and then either lowercase a-z or 26 more graphic characters. There are no accented characters.

To build a game with Ozmoo with accented characters, we typically need to:

- Decide which (less important) characters in PETSCII and in the C64 character set will be replaced with the accented characters we need.
- Create a font (character set) where these replacements have been made
- Create mappings from PETSCII to ZSCII for when the player enters text
- Create mappings from ZSCII to PETSCII for when the game outputs text

There are already fonts in place for some languages (see documentation/fonts.txt). You can also supply your own font.

The character mappings are created at the beginning of the file streams.asm. There are already mappings for the languages which there are fonts for (see the chapter on fonts in the Ozmoo manual).

To build a game using accented characters, the make command may look like this:

```
ruby make.rb examples\Aventyr.z5 -f fonts\sv\PXLfont-rf-SV.fnt -cm:sv
```

where

-f sets the font to use.

-cm sets the character mapping to use.

If you want to create a character mapping for say Czech, and you know you will never want to build a game in Swedish, you can just replace the Swedish mapping in streams.asm and use -cm:sv to refer to your Czech mapping.

The definition of ZSCII can be found at: <https://www.inform-fiction.org/zmachine/standards/z1point1/sect03.html#eight>

The accented characters which are available by default, and which Ozmoos can use, are in a table under 3.8.7.

The definition of PETSCII can be found at: <http://sta.c64.org/cbm64petkey.html>

Please read the notes below the table as well.

Note that in the mapping from PETSCII to ZSCII, you must also convert any accented characters from uppercase to lowercase, i.e. for Swedish PETSCII Å is mapped to a ZSCII ä.

If you build a game using Ozmoos's Debug mode (Uncomment the 'DEBUG' line near the start of make.rb), Ozmoos will print the hexadecimal ZSCII codes for all characters which it can't print. Thus, to create mappings for a game in a new language, you can start by running it in Debug mode to see the ZSCII character codes in use.

Assembly Flags

The flags described here can be set on the Acme commandline using the syntax
`-D[flag]=1`

General flags

`BGCOL=n`
`BGCOLDM=n`

Set the background colour for normal mode and darkmode.

`BORDERCOL=n`
`BORDERCOLDM=n`

Set the border colour for normal mode and darkmode.

`CACHE_PAGES=n`

Set the minimum number of memory pages to use for the cache (used to buffer pages otherwise residing at `$D000-$FFFF`).

`CHECK_ERRORS`

Enable some extra checks for runtime errors, making the interpreter a little bigger.

`CONF_TRK=n`

Point out the track on disk where the interpreter must look for the configuration information.

CURSORCOL=**n**
CURSORCOLDM=**n**

Set the cursor colour for normal mode and darkmode.

COL2=**n**
COL3=**n**
COL4=**n**
COL5=**n**
COL6=**n**
COL7=**n**
COL8=**n**
COL9=**n**

Replace color in Z-machine palette with a certain colour from the C64 palette.

CONF_TRK=**n**

Tell the interpreter on which track the config blocks are located (in sector 0 and 1).

CURSORCHAR=**n**

Use this character code as the cursor.

CUSTOM_FONT

Tell the interpreter that a custom font will be used.

FGCOL=**n**
FGCOLDM=**n**

Set the foreground colour for normal mode and darkmode.

FREE_SAVE_BLOCKS=**n**

Set the number of blocks the interpreter should expect to be available on a blank save disk. This number is used to calculate the number of save slots that can fit.

LURKING_HORROR

If the game uses sound, enable the code that defines assumptions we need to make for the Lurking Horror, i.e. use a hardcoded table defining which sounds have repetition enabled, as the Z-code opcode to play sound doesn't support setting this parameter in z3.

MAJOR_VERSION_NO

The major version number of the interpreter.

MINOR_VERSION_NO

The minor version number of the interpreter.

NOSCROLLBACK

Don't include the scrollbar buffer feature.

NOSECTORPRELOAD

Don't load any parts of static memory into RAM from disk sectors on boot. If make.rb can decide that this won't be needed anyway, it will set this flag, to make the interpreter a little smaller.

OPTIMIZE_VMEM

Enable continuous virtual memory optimization, moving vmem blocks around to make sure the most used blocks are in unbanked memory. This can only be used on C64 and C128, and has no effect when the REU Boost virtual memory model is activated, i.e. when the interpreter has been built with the REUBOOST flag and the player chooses to use an REU for caching game data.

PREOPT

Build the interpreter in PREOPT mode (used to pick which virtual memory blocks should be preloaded into memory when the game starts).

REUBOOST

Add support for an alternate virtual memory model which is simpler and faster, for use with REU. When the story file is stored in an REU, copying a page from REU to RAM is so fast that it doesn't pay off to spend a lot of time figuring out which vmem block to replace. Defining REUBOOST typically makes games played with an REU about 10% faster.

SCROLLBACK_RAM_PAGES=n

This value is typically between 24 and 48. Reserve this many pages at the top of RAM bank 0 for a scrollback buffer. However, if an REU is detected and used for scrollback at runtime, this RAM buffer should not be allocated. If the interpreter is built with REUBOOST, and there is room for the story file in the REU, and the user chooses to load the story file into the REU, this won't matter, because the top 12 KB of RAM bank 0 are unused in this scenario anyway. SCROLLBACK_RAM_PAGES is not used or supported for MEGA65, since it always uses AtticRAM for scrollback buffer.

SLOW

Prioritise small size over speed, making the interpreter smaller and slower. For X16, C128 and Plus/4, this is enabled by default and can't be disabled, due to their memory models.

SOUND (MEGA65 only)

Include support for sound (.wav sample files playback).

SPLASHWAIT=n

Set how many seconds the splash screen should show. If this is set to 0, disable the splash screen completely, making the binary slightly smaller. On the C64, the splash screen resides in the virtual memory buffer, which means story_start and the maximum amount of dynamic memory aren't affected for a virtual memory build, but the empty space does allow Exomizer to compress the boot file better.

STACK_PAGES=n

Set the number of memory pages to use for stack. Can't be less than 2 for a mode P build, or less than 4 for any other build, since the stack holds initialization code on boot. 4 is good for pretty much any ZIL or Inform 6 game. Inform 7 games expect a 64 page stack, and are likely to crash at some point if they get a stack as small as 4 pages.

STATCOL=n

Set the statusline colour. (only for z1-z3).

TARGET_C64
TARGET_X16
TARGET_C128
TARGET_PLUS4
TARGET_MEGA65

Set the target platform.

TERPNO=*n*

Set the interpreter number (0-19) as reported by the interpreter to the game. Default is 2 (Apple II) for Beyond Zork and 8 (C64) for other games.

TRACE

Allocate a page of memory to keep a record of the last 64 instructions that were executed. If both TRACE and DEBUG are defined, and the game crashes with a fatal error, the game prints out the ten last instructions that were executed (Z-machine address and opcode).

UNDO

Include support for Undo. Unless UNDO_RAM is also defined, this uses REU on C64 and C128, video RAM on Commander X16 (banked RAM in a version 6 build with pictures, where the tile store takes the video RAM over), and Attic RAM on MEGA65. Undo is not available on Commodore Plus/4.

UNDO_RAM

Include support for Undo in RAM (only on C128).

USE_BLINKING_CURSOR=*n*

Make the cursor blink during player input, where *n* is the number of jiffies between blinks.

USE_HISTORY=*n*

Enable command history, and allocate at least *n* bytes for storing it. Since the code to support command history is small, and since the storage is placed between the interpreter and the next thing in RAM (virtual memory buffers or stack or virtual memory) which needs to be page-aligned anyway, this can sometimes be enabled without causing `story_start` to move up.

VMEM

Utilize virtual memory. Without this, the complete game must fit in C64 RAM available above the interpreter, all in all about 50 KB. Also check section “Virtual memory flags” below.

X_FOR_EXAMINE

Enable support for automatically changing the word “X” to “EXAMINE”, when it’s typed as a verb by the player.

Z1
Z2
Z3
Z4
Z5
Z6
Z7
Z8

Build the interpreter to run Z-machine version 1, 2, 3, 4, 5, 6, 7 or 8.

Z6_ECM_MODE

C64 and version 6 only. Use the VIC-II’s Extended Color Mode, so that each of the eight windows can have its own background colour. Set by make.rb’s -ecm switch. See “Version 6”.

Z6_FCM_MODE

MEGA65 and version 6 only. Use the VIC-IV’s Full Colour Mode, with 16-bit character codes, giving an 80x25 screen on a 640x200 (H640) canvas, or the legacy 320x200, 40x25 one with -fcm:40. Set by make.rb’s -fcm switch. See “Version 6”.

Z6_PICTURES

Version 6 only, and only on the MEGA65 (where it requires Z6_FCM_MODE) and the X16 (where there is nothing else to ask for). The game’s pictures are built alongside it and are drawn: on their own picture disks, preloaded into Attic RAM, on the MEGA65; in the game’s directory, read from the SD card as they are drawn, on the X16. Set by make.rb’s -pics switch. See “Version 6”.

PIC_STAGING_BANK=n

X16 and Z6_PICTURES only. The first bank of banked RAM above the story, where the picture being drawn is staged (four banks) and, above that, the undo buffer. Computed and passed in by make.rb, which knows the story's size.

DANISH_CHARS
FRENCH_CHARS
GERMAN_CHARS
ITALIAN_CHARS
SPANISH_CHARS
SWEDISH_CHARS

Map national characters in ZSCII to their PETSCII equivalents. No more than one of these can be enabled.

Internal Flags

These flags are set and used internally in Ozmoo, depending on the settings of the general flags, especially the TARGET_??? flag.

COMPLEX_MEMORY

If COMPLEX_MEMORY is set, then the Z-machine dynamic memory is not directly accessible in RAM, so we can't use simple lda/sta to access it. Instead, special routines are required. For C128 and X16, this involves selecting the right RAM bank, for MEGA65 it involves far memory access, while for Plus/4 it just means we need to bank out ROM which normally covers large areas of RAM during execution. Plus/4 has COMPLEX_MEMORY set, but not FAR_DYNMEM.

FAR_DYNMEM

If FAR_DYNMEM is set, then the Z-machine dynamic memory is not in the part of RAM which can be directly accessed with 16-bit addresses. This means we can't use simple lda/sta operations, and we can't assume that dynamic memory starts at the story_start marker. Thus, additional code is needed to read and write to dynamic memory. A platform that has FAR_DYNMEM set must also have COMPLEX_MEMORY set.

HAS_SID

If the target has a SID chip for sound generation.

SUPPORT_80COL

If the target has an 80 column display.

SUPPORT_REU

If the target potentially supports extended memory of some sort, to be used as a RAM disk for faster game play.

VMEM_END_PAGE=n

This is the page where virtual memory storage ends, i.e. the last page that can be used by the virtual memory system plus one. For the C64, the last page is \$FF (address *FF00–FFFF*), so **VMEM_END_PAGE** is \$00.

Z6_MOUSE

Set when the build has a screen that can show a mouse pointer: a MEGA65 build with **Z6_FCM_MODE**, or an X16 build with **Z6_PICTURES**. It guards *asm/mouse.asm* and everything that depends on a mouse existing — the **Flags 2** mouse bit, the *read_mouse* and *mouse_window* opcodes, and the mouse clicks in the default set of terminating characters. See “Mouse”.

Debug flags

(If any of the flags in this section are enabled, **DEBUG** is automatically enabled too.)

DEBUG

Print debug information, print descriptive error messages, store a trace of the instructions which have been executed and print them in case of an exception. Also check section “Debug flags” below.

BENCHMARK

When the game starts, replay a number of colon-separated commands which have been stored with the interpreter (in the file *text.asm*). Charcode 255 in this command sequence means print the number of jiffies since the computer was started.

COUNT_SWAPS

Keep track of how many vmem block reads have been done.

PRINT_SWAPS

Print information whenever a memory block is loaded into memory.

TRACE_FLOPPY

Print trace information for floppy operations.

TRACE_FLOPPY_VERBOSE

Print verbose trace information for floppy operations.

TRACE_PRINT_ARRAYS

Print the contents of the input array and the parse array whenever a read command has been performed.

TRACE_READTEXT

Print trace information for opcode read/aread/sread.

TRACE_SHOW_DICT_ENTRIES

Print which dictionary words are checked when trying to match a word, and if the input word was less than or greater than that word.

TRACE_TOKENISE

Print trace information for the tokenization process.

TRACE_VM

Print trace information for the virtual memory system.

VICE_TRACE

Send the last instructions executed to Vice, to aid in debugging.

VIEW_STACK_RECORDS

Print whenever the stack size hits a new high, or when the number of bytes pushed onto the stack within a frame reaches a new high.

Configuration blocks

Interpreter configuration is stored in two blocks on the boot disk (track 1, sector 0-1). The interpreter loads this information when it starts.

Contents:

4 bytes: Game signature: A 32-bit random number.

--- Disk information block ---

1 byte: Number of bytes used for disk information, including this byte

1 byte: Interleave (in range 1-21)

1 byte: Number of save slots which can fit on a disk (in range 2-132)

1 byte: Number of disks used (disk 0 is the save disk(s),
disk 1 .. are game disks

For each disk:

1 byte: n: Number of bytes used for this entry, including this byte.

1 byte: Device#. Should be 8,9,10,11 or 0, meaning it's not decided
yet - it's up to the terp to set the device#.

2 bytes: Last story block # + 1 (high endian) on this disk

1 byte: t: Number of tracks used for story data

(If this number is > 0, this is a story disk!)

t bytes: Sectors used for story data:

Bit 0-4: # of sectors used.

Bit 5-7: First sector# used.

(Example: %01010000: Use 16 sectors, starting with #2)

x bytes: Disk name in PetSCII. Special codes:

0: End of string

128: "Boot "

129: "Story "

130: "Save "

131: "disk "

--- Vmem information block ---

2 bytes: Number of bytes used for vmem information, including this byte (high endian)

1 byte: Number of vmem blocks suggested for initial caching

1 byte: Number of vmem blocks already preloaded
x bytes: vmap_z_h contents for the suggested blocks
x bytes: vmap_z_l contents for the suggested blocks

Typical size for a 3-disk (+ save disk) game:

$4 + 1 + 1 + 1 + 1 + 2 * (1 + 1 + 2 + 1 + 40 + 5) + 2 * (1 + 1 + 2 + 1 + 0 + 3) + 1 + 1 + 1 + 100 + 100 = 8 + 2 * 50 + 2 * 8 + 203 = 8 + 100 + 16 + 203 = 327$ bytes

An interpreter needs to reserve this space for disk information:

z1/z2/z3: $1 + 1 + 1 + 1 + 2 * (1 + 1 + 2 + 1 + 0 + 3) + (1 + 1 + 2 + 1 + 40 + 6) = 4 + 2 * 8 + 51 = 71$ bytes

z4/z5/z8: $1 + 1 + 1 + 1 + 2 * (1 + 1 + 2 + 1 + 0 + 3) + 2 * (1 + 1 + 2 + 1 + 40 + 5) = 4 + 2 * 8 + 2 * 50 = 120$ bytes (* see Double 1571 drive support below)

1571 drive support:

- z1/z2/z3 games: No change
- z4/z5 games: A single disk using only track 1-53 can hold all story data. Disk information will then fit in less than the number of bytes stated above.
- z8 games: Disk information for a single drive game will fit in the number of bytes stated above.

Double 1571 drive support:

- z1/z2/z3/z4/z5 games: Any game fits in less than the amount of bytes calculated for a 3-disk game above.
- z7/z8 games: We need an additional 28 bytes to fit a full-size game. Thus, we allocate 150 bytes, to have some margin.

1581 drive support: This system should work, without extending the limits above, using only 31 sectors per track for story data, while allowing z8 games of up to 512 KB in size on a single disk. If we want to store several games on a single 1581 disk, we should add a track offset in each disk entry (i.e. saying that tracks below track x are considered empty).

Printer Support

Currently Ozmoos doesn't have any support for printers, but hooks exist that could be used to add such functionality.

Infocom provided printer output through the transcript command, which redirected text output to output stream 2. `stream.asm` implements streaming, and `stream__print__output` can be extended to handle stream 2 and send the bytes to the printer.

We currently have no hardware to test printer support on, and no immediate plans to provide such support. However, we are happy to accept patches.

Runtime Errors

These are the runtime errors that may occur when running Ozmoo:

- `ERROR_UNSUPPORTED_STREAM = 1`

The Z-machine supports certain streams for input and output of text. If a program tries to open a stream number which is not defined or which is not supported by Ozmoo, this error occurs.

- `ERROR_CONFIG = 2`

The boot disk of a program built with Ozmoo has config information on sector 1:0 and 1:1 (unless it's built as a single file program). This error means there seems to be something wrong with this information. The information is copied to memory when the game boots and then used whenever a block of game data needs to be copied from disk, so the problems may be discovered when the boot disk is no longer in the drive.

- `ERROR_STREAM_NESTING_ERROR = 3`

The Z-machine has a stack for memory streams. If the program tries to pull items from this stack when it is empty, this is the resulting error.

- `ERROR_FLOPPY_READ_ERROR = 4`

There was a problem reading from the disk.

- `ERROR_STACK_FULL = 6`

The program tried to make a routine call or push data onto the stack when there was not enough room. This means there is a bug in the program, or it needs a bigger stack. Stack size can be set with a commandline parameter to `make.rb`.

- `ERROR_STACK_EMPTY = 7`

The program tried to pull a value from the stack when it was empty.

- `ERROR_OPCODE_NOT_IMPLEMENTED = 8`

An unknown Z-machine opcode was encountered. This can happen if the wrong disk is in the drive when Ozmoos tries to retrieve a block of program code.

- `ERROR_USED_NONEXISTENT_LOCAL_VAR = 9`

Each routine in Z-code has between 0 and 15 local variables. If an instruction references a local variable number which is not present in this routine, this is what happens. Normally, a compiler like Inform doesn't let the programmer write code which can cause this, unless you skip the highlevel language and write Z-machine assembler.

- `ERROR_BAD_PROPERTY_LENGTH = 10`

The program tried to use an object property of an illegal length, where a property value has to be one or two bytes.

- `ERROR_UNSUPPORTED_STORY_VERSION = 11`

The first byte of the story file must match the Z-machine version for which the interpreter was built (3, 4, 5 or 8). Otherwise, this occurs.

- `ERROR_OUT_OF_MEMORY = 12`

The program referenced memory which is higher than the last address in the story file.

- `ERROR_WRITE_ABOVE_DYNMEM = 13`

The program tried to write to memory which is not part of dynamic (RAM) memory.

- `ERROR_TOO_MANY_TERMINATORS = 15`

A dictionary in a Z-machine program holds a list of terminating characters, which are used to separate words. It is illegal for this list to hold more than ten characters. If it does, this error occurs.

- `ERROR_NO_VMEM_INDEX = 16`

The `vmem_oldest_index` is only populated inside the loop if we find a non-PC vmem block older than the initial `vmem_oldest_age` of \$ff. That should always happen, but to assert that the code is correct the debug version of Ozmoos checks that `vmem_oldest_index` is valid and issues `ERROR_NO_VMEM_INDEX` if not.

- `ERROR_DIVISION_BY_ZERO = 17`

An attempt was made to divide a number by zero.